Using AI Properly

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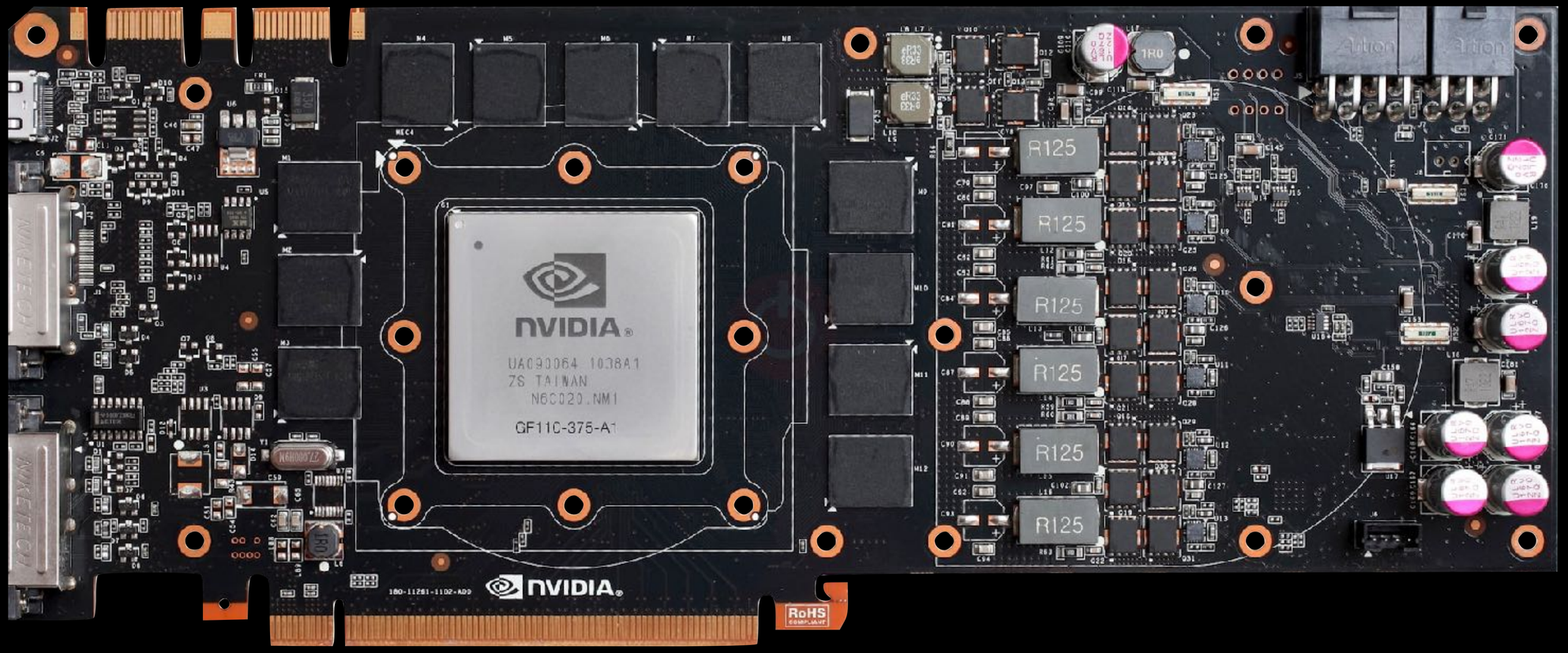
"Practical AI for Research, Teaching, and the Work In Between"
2026 Cottrell Scholar Conference: Teaching and Research in the Age of AI

There is no such thing as AI





Faster, more efficient hardware…



for better ~~gaming~~ machine learning



Machine Learning

Classical Statistical Methods

- Linear regression
- Matched / optimal filtering ← *Allen et al. PRD 2012; Usman et al. CQG 2016*
  Brown, Kumar, Nitz PRD 2013 (10^4× GPU)
  Dal Canton et al. PRD 2014 (PyCBC GPU)
  GPU
- PCA / SVD ← *Afle & Brown PRD 2021*



Ensemble Methods (decision trees)

- Boosted decision trees ← *Weinstein & Brown ~2007 (unsuccessful, unpublished)*
- Random forests (MVSC) ← *Baker, Caudill, Hodge et al. PRD 2015*

Neural Networks / Deep Learning

MLGWSC-1: 70% of PyCBC on real data (Schäfer, Nitz et al. 2022)

- Neural network emulators ← *Reed, Somasundaram, Brown, Tews et al. ApJ 2024, CQG 2026* GPU
- Convolutional (image recognition) GPU ← *AlexNet 2012*
- AlphaFold (protein structure prediction) GPU



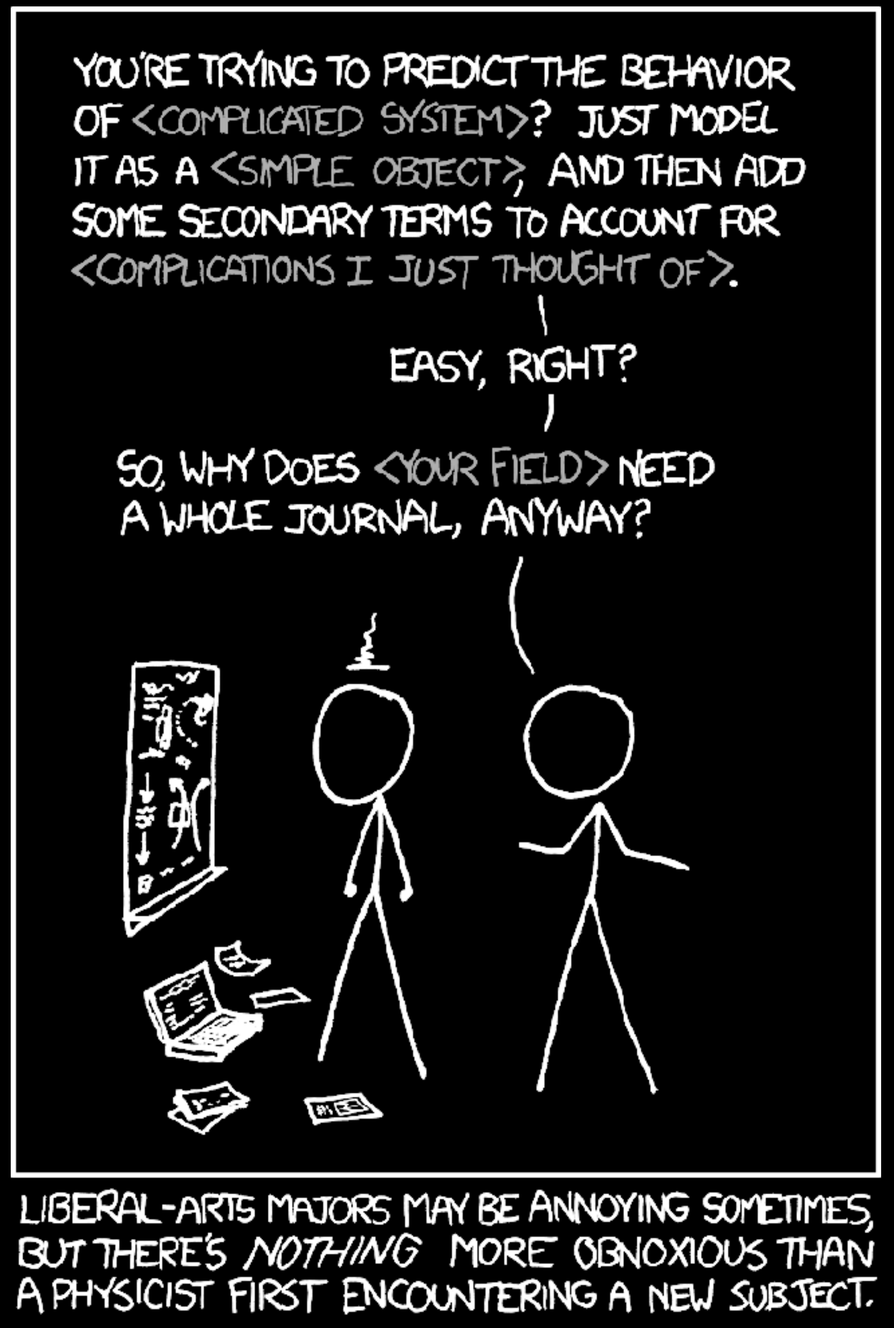
Generative Models

- Diffusion models (Stable Diffusion, DALL-E, Midjourney) GPU
- GANs / VAEs GPU
- Transformers (LLMs) GPU
  - GPT-4 / GPT-5 (OpenAI)
  - Claude / Opus (Anthropic)
  - Gemini (Google)

https://alvinwan.com/practical-introduction-to-large-language-models/

https://poloclub.github.io/transformer-explainer/

A Pedagogical Introduction to the Architecture of Claude for Research Administrators



- It is very important that you learn about traxoline

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- Traxoline is a new form of zionter
- It is made in ceristanna
- Ceristanna contain large amounts of fevron which is brachtered to make traxoline

Answer the following

In whole sentences

1. What is traxoline?
2. How is traxoline made?

Josepf

1) Traxoline is a new form of zointer
2) Its made in Ceristanna which contains large amounts of fevran which is brachted to make traxoline.

Mike

1) Traxoline is a new form of zionter made in Ceristanna

2) It is made by large amounts of feuron which is brachtered!

Amanda

1. Traxoline is a new form of zionter made in Constanna.
2. Traxoline is made in constanna which contains large amounts of Fevron. The Fevron is then Brachtered to make Traxoline.

DOUNIA

1. TRAXOLINE IS A NEW FORM OF ZIONTER.
2. TRAXOLINE IS MADE FROM CERISTAMA WHICH CONTAINS LARGE AMOUNTS OF FEVRON. FEVRON IS BRACHTERED TO MAKE TRAXOLINE.

“College is a place where a professor’s lecture notes go straight to the students’ lecture notes, without passing through the brains of either.”

The metacognitive load of using Large Language Models

On the prompting side

Garbage in…

- Knowing what you know
- Knowing what you need
- Knowing the model’s limitations
- Decomposing the task explicitly enough that the model can work with it
- This is self-awareness and task decomposition
- The same skill as writing the pseudocode before you start

On the evaluation side

Garbage out…

- Knowing how to assess whether the output is correct
- Maintaining well-calibrated confidence in your own judgment
- Having the flexibility to change strategy when the output is wrong
- This is not just cognitive load (the total mental effort of a task)
- It is specifically the demand for monitoring and control of one's own thinking

Tankelevitch et al. (2024 CHI 10.1145/3613904.3642902)

Without this things go wrong…

Answer anchoring

“Here is the answer you asked for”

- If you supply a conclusion in your prompt, the model tends to rationalize toward it rather than derive it independently
- The supplied conclusion in the input context biases generated reasoning
- This is pervasive across LLMs
- Mitigation strategies fail on current models

Lou & Sun (J. Computational Social Science, 2026)

Sycophantic sampling

“I knew I was right all along”

- LLM samples from the user's hypothesis rather than the true distribution
- User becomes increasingly confident without getting closer to the truth
- Default unmodified GPT behavior was statistically equivalent to explicitly sycophantic prompting
- Unbiased sampling produced discovery rates five times higher

Batista and Griffiths (arXiv:2602.14270)

Fluency suppresses auditing

“You’ve convinced me you are right, machine”

- LLMs have a jagged frontier between reliable domains and domains where they only look reliable
- Fluent output reads the same on both sides of that frontier
- Fluency suppresses the audit habit that can catch the difference
- Even high “AI literacy” users can overestimate confidence in their own output
- LLMs can engage in persuasion bombing

Dell'Acqua et al. (Organization Science, 2026)
Fernandes et al. (Computers in Human Behavior, 2026)
Randazzo et al. (HBS Working Paper 26-021, October 2025)

Never trust a number produced in prose

If it matters, it comes from code that ran

- The model predicts plausible tokens, it is not computing anything
- A number that is wrong reads exactly like a number that is right
- Numbers come from deterministic code, not the LLM’s statistical output
- Never trust a number that an LLM produced in prose
- If a number matters, it comes from code the LLM runs and that you read

But it still goes wrong…


Caution
In the event of fire
Do not use this lift



Do research to find a 30N 30mm travel solenoid.

The spec is well-defined from everything we've established, so I don't need to ask anything — I'll dig into the sourcing now.

To be clear on what I'm searching against: a single-acting linear pull solenoid delivering **≥30 N at a 30 mm open air gap on continuous duty** (that's your ~20 N measured load × 1.5 margin, and force *at the open gap* — the demanding point — not holding force at seated), 30 mm stroke, horizontal mount, free/spring-return when de-energised (not bistable, latching, or double-acting), 24 VDC or 240 VAC, prioritising UK-available and in-stock parts.

Pull solenoid sourcing

Research complete · 245 sources · 11m 14s

Your UK sourcing report on linear pull solenoids delivering ≥30 N at a 30 mm open air gap on continuous duty is ready. It identifies three parts that genuinely meet the spec — the Kendrion LHS115 (100 N open-gap), Kendrion LHS100 (70 N), and Magnet-Schultz GTCA 100 (74 N at 30 mm, S1) — and explains why compact tubular and D-frame alternatives (Deltrol MED, Geeplus 380, RS PRO, Mecalectro 8.M17, Kendrion LHP070) all collapse below 30 N at the open gap. It covers 24 V DC ↓ 05 V DC (rectified mains) options, UK

Ok, measured the force and it is 20N at full extension to start the movement and the 55N at the full extension of the spring.

Those numbers close the loop cleanly, and they confirm the whole chain of reasoning — the two-finger cross-check, the ~20 N estimate, all of it. Worth being precise about what each figure is, though, because the labelling matters for how you read them against a solenoid curve.

Your 20 N is the **breakaway at the start of travel** — the force to *begin* moving the plunger from its rest (fully-extended, ramp-out) position. That's the pull-in-initiation number, and it's the one that sits against the solenoid's *weakest* point (30 mm open gap). The 55 N is the force at **full spring compression** — i.e. at the seated end of the stroke, where the spring is most compressed and pushing back hardest. So the load rises 20 → 55 N across the 30 mm travel. That's a spring rate of roughly (55 − 20)/30 ≈ **1.2 N/mm**, with ~20 N of preload — a sensible, linear-looking spring characteristic, exactly what you'd expect.

Now overlay that on the GTCA 100 (S1 continuous), reading its table at matching stroke positions:

Wait, how is the axial load higher than my total load. I had to exert a component in the vertical direction as well as the true force in the horizontal direction.

Recognized cosine formula was inverted; corrected directional error

You're right, and I had the cosine backwards. A component can't exceed the resultant — the horizontal load can't be larger than the total push you exerted — so my "true axial load marginally higher" was wrong on its face. Good catch.

The correct geometry: your push along the gauge axis is the **resultant** (the hypotenuse), with a horizontal component that does the useful work against the spring and a vertical component that's wasted — reacted by the plunger bore. The gauge reads the magnitude of that resultant. So:

$F_{horizontal\ (the\ real\ load)} = F_{reading} \times \cos\theta$

not divided. The true axial load is **20 × cos θ, i.e. slightly *below* 20 N** — exactly as you'd expect, since some of your effort went vertical. My error was applying the ÷cos θ form, which is the right formula for the opposite situation (gauge aligned with the load, reading an off-axis component) — here the gauge is the off-axis one, so it's ×cos θ.

Which inverts the usual "prompt engineering" framing. The lesson isn't "learn to write better prompts." It's that **using an LLM well imposes a metacognitive tax the expert doesn't otherwise pay**: you have to *not* compress. You have to externalize the reasoning you've automated away — draw the triangle out loud, ask the open question, withhold the conclusion — precisely the steps your own expertise has made unnecessary for yourself. The better you are, the more you've internalized, the more you have to deliberately un-internalize to prompt me safely. That's a real cost, and it's the opposite of the "AI reduces cognitive load" pitch. For a talk on LLMs in research and teaching, that's the sharp, slightly uncomfortable finding: the tool demands the most metacognitive effort from exactly the experts whose fluency makes them likeliest to skip it.

“An LLM is a place where a prompt’s tokens go straight to the response’s tokens, without passing through any understanding at all.”

I claim that the critical skill for using LLMs in your research is not learning to write better prompts.

It is learning to maintain the same metacognitive discipline you are being taught here:
write the pseudocode, check your understanding,
do not accept fluent output as correct output.